

ER Modeling: Objects, Properties and Rules

A college records enrollments for Fall 2026. S101 is a student; DB101 / 1 is class 1 of course DB101.

Question	One concrete answer
Which objects?	Student S101; course DB101; section DB101 / 1
Which properties?	S101 has a name and recorded phone numbers.
Which connections?	S101 enrolls in DB101 / 1.
Which rules?	Every section belongs to exactly one course.

1. Miniworld: the part of the world represented by the database.

2. Scope: our small campus example covers Fall 2026 only.

3. Illustrative data: these are invented facts, not university policy or SQL output.

Chapter order: entities and attributes; relationships; weak entities; library case; ternary relationships; UNIVERSITY case.

Remember: An ER model describes objects, properties, connections and rules.

Entities, Entity Types and Entity Sets

Add student S104. Do we need a new STUDENT type?

STUDENT describes	Properties
An entity type	StudentId, Name, Phone

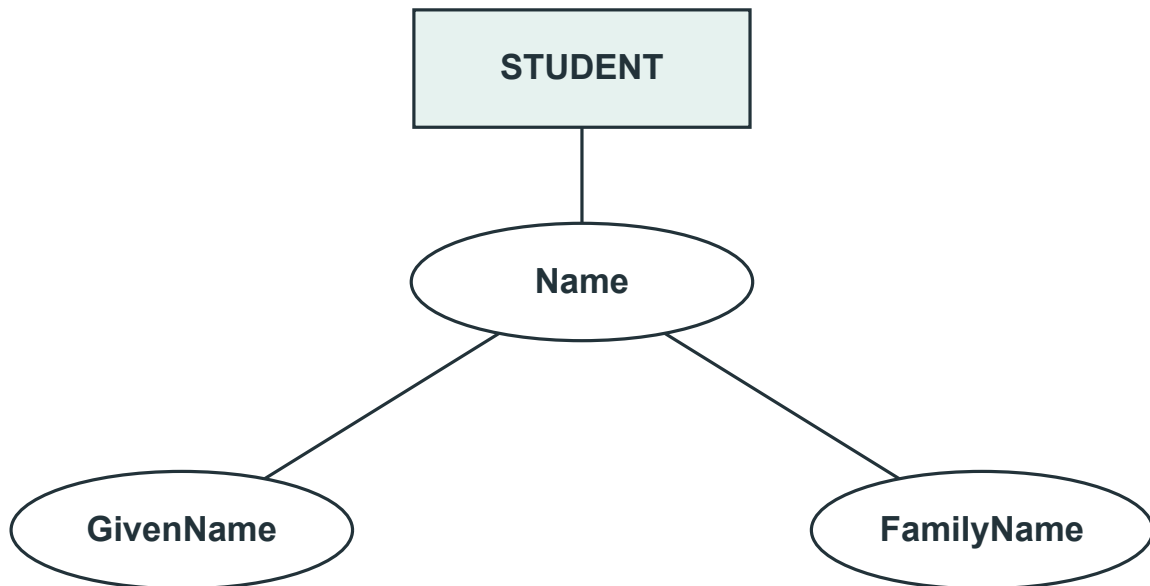
StudentId	Name	Phone values
S101	Alex Lin	02-0000-0101 02-0000-0102
S102	Blair Wu	02-0000-0201
S103	Casey Chen	None recorded

- 1. Entity:** one student, identified here by S101.
- 2. Entity type:** STUDENT and its shared properties.
- 3. Entity set:** the students currently present. A person is not the identifier string.

Remember: Adding S104 changes the current set, not the STUDENT definition.

Attributes: Simple and Composite

We must search family name Lin separately. Which parts should Name have?



ER diagram: an oval describes a property; connecting ovals show components.

Student	GivenName	FamilyName
S101	Alex	Lin

- 1. Name is composite:** GivenName and FamilyName are its parts.
- 2. Simple attribute:** no further components are used by this model.
- 3. Change the requirement:** only display Alex Lin as a whole. Name may remain one simple attribute; the component ovals are unnecessary.

Remember: A composite attribute has meaningful components in this model.

Attributes: Single-Valued and Multivalued

S101 has two phones. Do we draw two students or a double oval?



ER diagram: Phone is a multivalued attribute of STUDENT.

Student	Recorded phones	How many values?
S101	02-0000-0101; 02-0000-0102	2
S102	02-0000-0201	1
S103	None recorded	0

- 1. Single-valued:** at most one value when known.
- 2. Multivalued:** several values may belong to one entity.
- 3. Answer:** one S101, two Phone values. The double oval preserves both without inventing a second student. This table is not a SQL storage design.

Remember: The double oval means multivalued, not composite.

Attributes: Stored and Derived

S101 is enrolled in two sections.



ER diagram: a dashed oval marks a derived attribute.

Student	Section enrolled in
S101	DB101 / 1
S101	CS102 / 1

Student	EnrollmentCount
S101	2

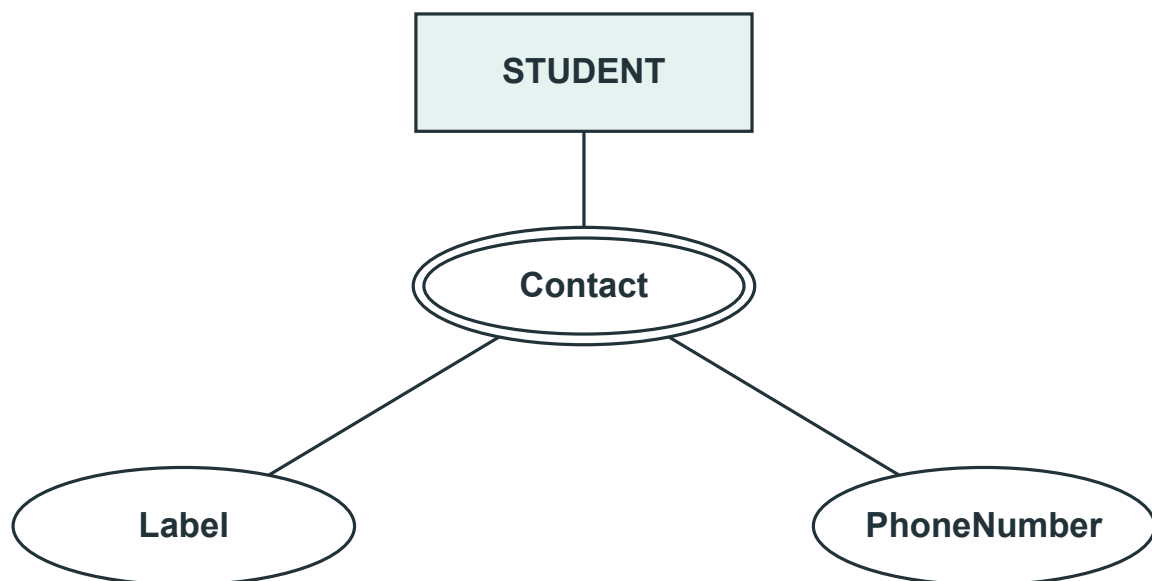
1. Stored: the first table records S101's two enrollments.

2. Derived: the second table shows their count, 2.

Remember: EnrollmentCount = 2 is calculated from S101's two enrollment records.

Attributes: Composite and Multivalued Together

Now the college must distinguish home from mobile numbers. Each Contact stores a Label and PhoneNumber, replacing the earlier bare Phone list.



ER diagram: Contact is both composite and multivalued, a complex attribute.

Student	Label	PhoneNumber
S101	home	02-0000-0101
S101	mobile	02-0000-0102

1. **One student; two Contact values; two components per value.**
2. **Single oval:** permits only one Contact value per student. S101 could not keep both home and mobile contacts.

Remember: Components and number of values answer different questions.

Keys: Identify One Entity

Two students are named Alex Lin. Would underlining Name reject valid data?

Proposed attribute	Declared rule	A key?
StudentId	No two students share it.	Yes
Name	Different students may share it.	No



ER diagram: the solid underline marks StudentId as a key of STUDENT.

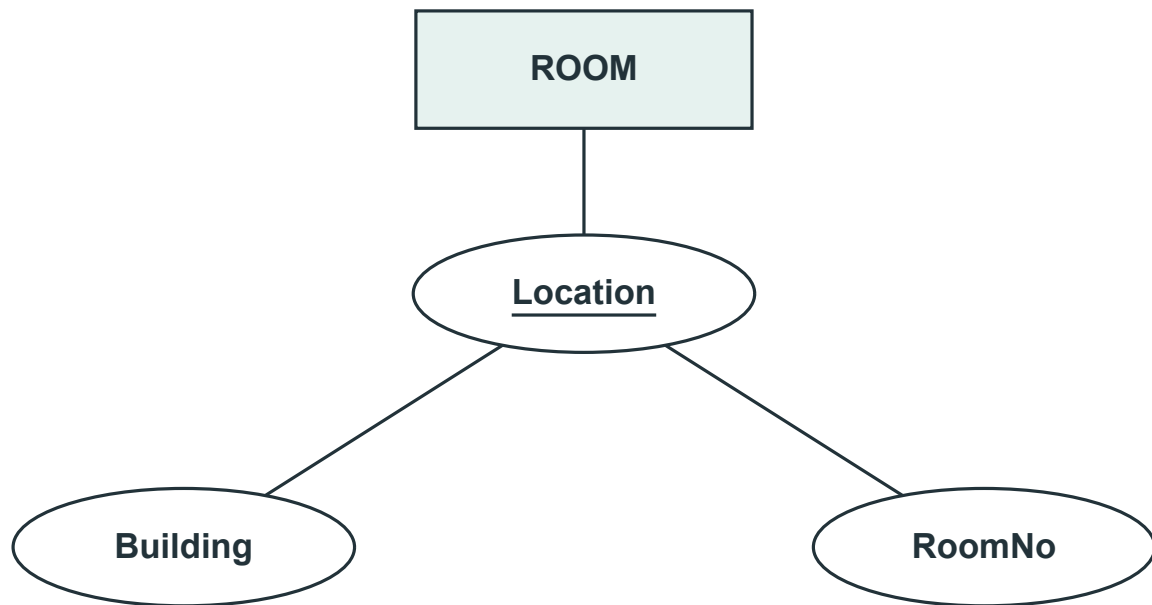
StudentId	Name
S101	Alex Lin
S104	Alex Lin

- 1. Solid underline:** each separately underlined attribute is an independent key.
- 2. Wrong claim: Name is a key.** These two allowed students share Alex Lin; StudentId distinguishes them.
- 3.** Choosing a relational primary key comes later, in mapping.

Remember: A key must identify an entity in every legal state.

Composite Keys: Building and Room Number

The campus has buildings A and B. A/101 means room 101 in building A. Can RoomNo 101 identify one room?



ER diagram: underline Location, the composite key, not each component separately.

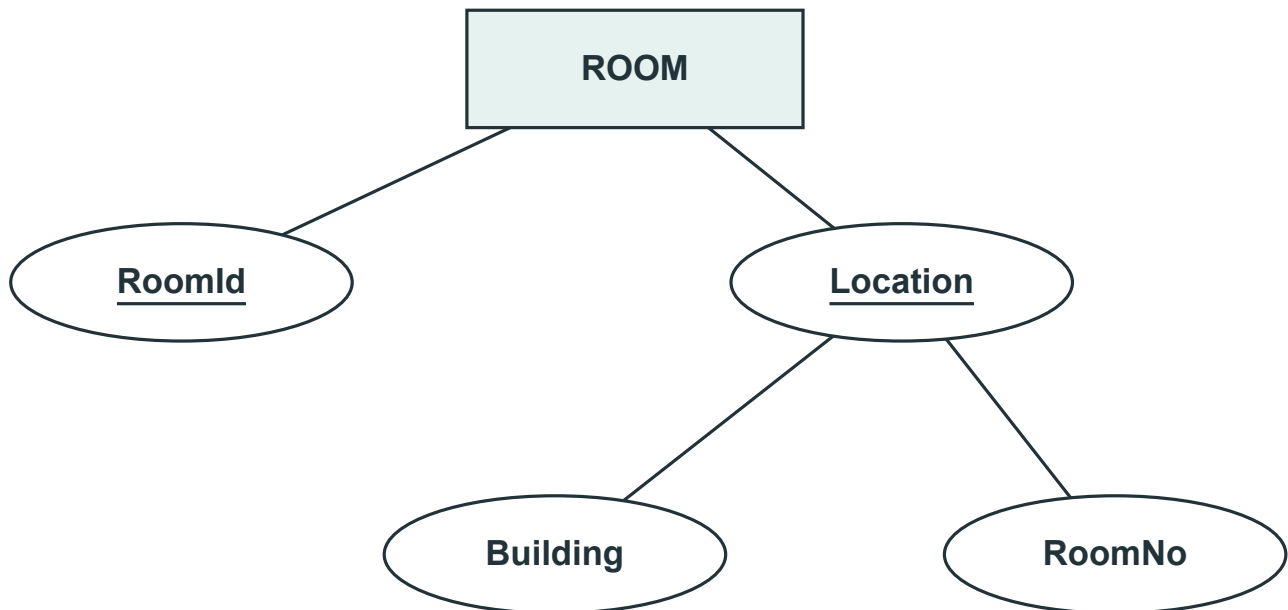
Information supplied	Matching rooms
RoomNo = 101	A/101 and B/101
Building = A	A/101 and A/102
A and 101	A/101 only

Declared rule: a room number is unique within a building. A minimal key needs both components; extra unnecessary attributes do not belong in it.

Remember: Building and RoomNo are needed together; neither identifies a room alone.

Keys: Two Independent Ways to Identify a Room

Same rooms, new rule: each also gets a unique RoomId. Must we supply all three values?



Two solid underlines mark two independent keys. Components are not independently underlined.

<u>RoomId</u>	<u>Building</u>	<u>RoomNo</u>
R1	A	101
R2	A	102
R3	B	101

- No:** R1 alone and A plus 101 each identify the same room.
- Building and RoomNo each repeat. Underlining either alone would incorrectly claim it is unique.

Remember: RoomId alone works; Location works through both of its components.

Domains: Allowed Attribute Values

Credits must be a whole number from 1 through 6. Can a new course have 5?

Current course	Credits
DB101	3
CS102	2

Proposed value	Check against the declared domain
5	Allowed: whole number, within 1-6
2.5	Rejected: not a whole number
7	Rejected: above 6
0	Rejected: below 1

1. Five is allowed even though neither current course uses it.
2. A value set, or domain, constrains possible attribute values.
3. This is an illustrative rule, not the university's credit policy.

Remember: A domain gives allowed values, not just currently observed values.

Missing Values: Unknown Is Not Zero

The college stores course grades and parking-permit numbers. Unfinished marking and no parking permit explain different missing values.

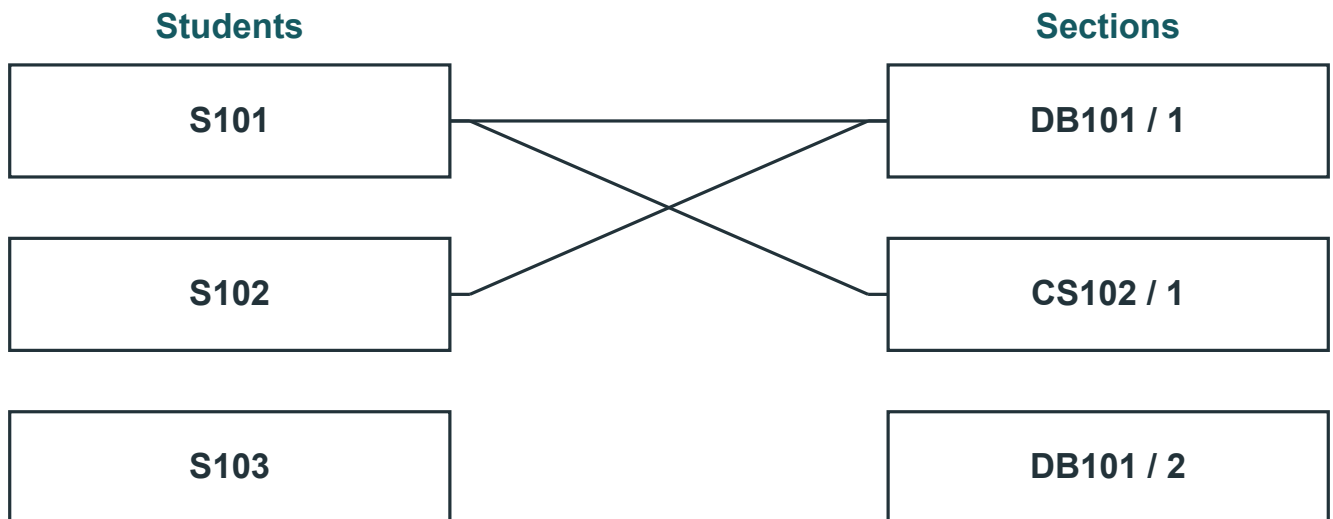
Known situation	What can we conclude?
Marking finished; recorded Grade = 0	The grade is known to be zero.
Marking is unfinished	The grade is unknown, not zero.
A person has no parking permit	A permit number is not applicable.
Nobody checked whether a permit exists	We do not know whether a number exists.

1. The textbook uses **NULL** for unknown or inapplicable information.
2. An empty entry alone does not explain which reason applies.
3. No recorded phone does not prove that the student owns no phone.

Remember: Unknown, inapplicable and a known zero have different meanings.

Relationships: Individual Enrollment Facts

Three students appear below. Are there also three enrollments? Follow each line.



Individual objects, not ER types. Each line is one enrollment; Grade is omitted here.

Inspect	Recorded participation
S101	2 lines
S103	0 lines
DB101 / 1	2 lines

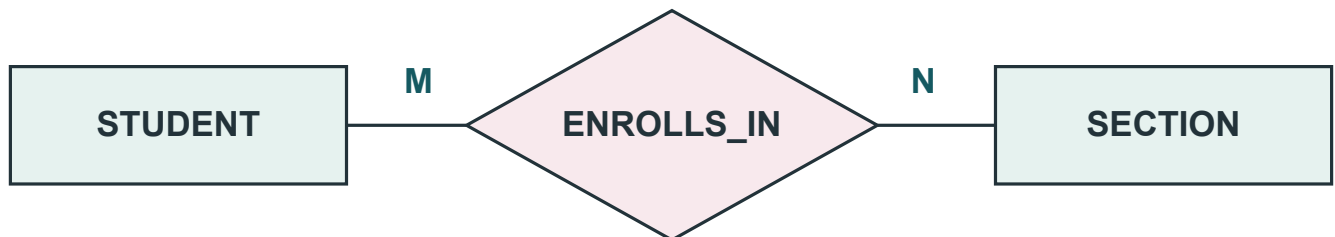
1. Relationship instance: one connection, such as S101 with DB101 / 1.

2. Relationship set: all three enrollment facts shown here.

Remember: S101 has two enrollments; S103 has none and can still exist.

Relationship Types: Reading the Diamond

ENROLLS_IN names the relationship type, not one student's enrollment.



ER relationship close-up. Keys and SECTION identification are omitted on this page.

How many sections can one student enroll in? Look at **N** beside SECTION.

How many students can enroll in one section? Look at **M** beside STUDENT.

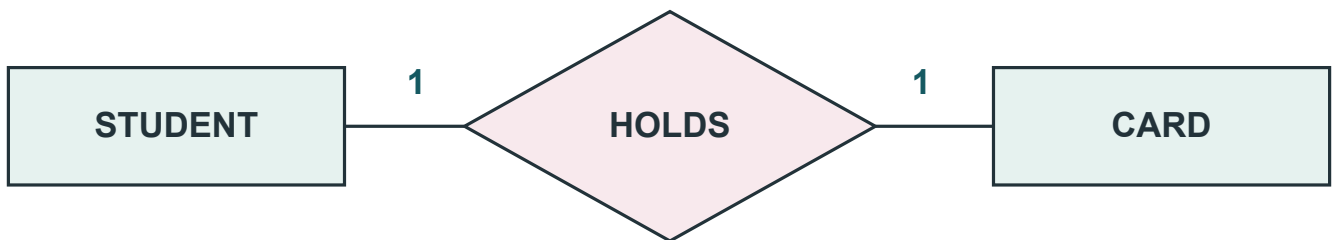
What are we counting?	Answer in the preceding example
Participating roles in each fact	2: student and section; binary
Recorded enrollment instances	3
Entity types connected by the diamond	STUDENT and SECTION

1. A rectangle names an **entity type**; a diamond names a **relationship type**.
2. Lines join participants. They are not arrows showing execution order.

Remember: Binary means two participating roles, not two recorded facts.

Cardinality Ratios: 1:1, 1:N and M:N

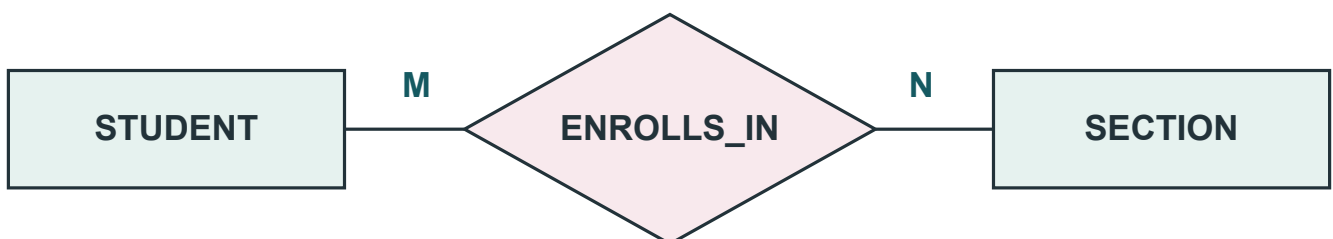
A campus records access cards, teaching and enrollment. K10/K11 are access-card IDs.



1:1 variant: at most one card per student; at most one student per card.



1:N: several sections per instructor; at most one instructor per section.



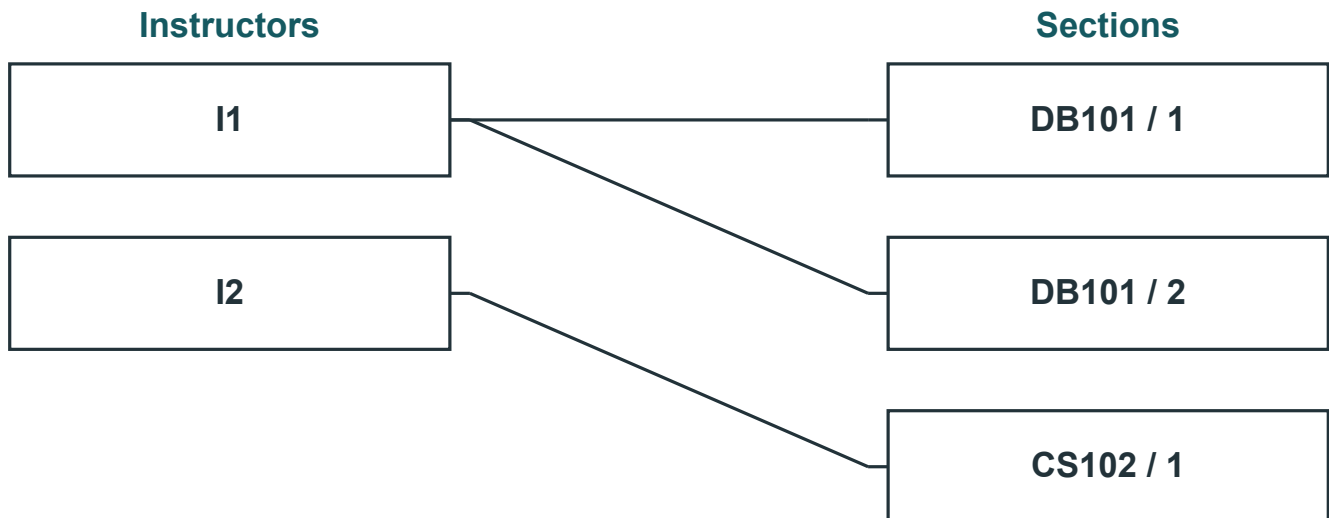
M:N: several sections per student; several students per section.

S101 cannot hold K10 and K11 in the 1:1 case. S101 and S102 may share DB101 / 1 in M:N.

Remember: 1:1, 1:N and M:N describe permitted maxima, not today's counts.

Cardinality: One Teacher, Many Sections

I1 and I2 are instructors. Each section has at most one teacher. Can I2 also teach DB101 / 1, which already has I1?



Each line is one teaching assignment between individual objects.



Maximum-ratio close-up; identification symbols are omitted.

1. The **1** near **INSTRUCTOR** limits instructors for one section.
2. The **N** near **SECTION** permits several sections for one instructor.

Remember: I1 teaching twice is allowed; two instructors for DB101 / 1 are not.

Participation: Partial and Total

Must every instructor teach? Must every section have an instructor?



Participation in TEACHES: single line at INSTRUCTOR; double line at SECTION. Keys omitted.

Side	Participation	Meaning and example
INSTRUCTOR	Partial (single line)	May teach no sections. I3 can have no teaching assignment.
SECTION	Total (double line)	Must have at least one instructor. A section with no instructor is not allowed.

1. Partial allows zero; it does not require anyone to teach zero sections.

2. SECTION's double line requires an instructor; the **1 near INSTRUCTOR** permits at most one.

3. Adding a second instructor is not allowed. Replacing one instructor with another is allowed.

Remember: An instructor may teach zero or more sections. Every section needs exactly one instructor.

Min-Max Notation: Read Beside Each Entity

I1 teaches two sections; I3 teaches none. Which adjacent pair permits both?



Each pair bounds one adjacent entity's participation count. Identification is omitted.

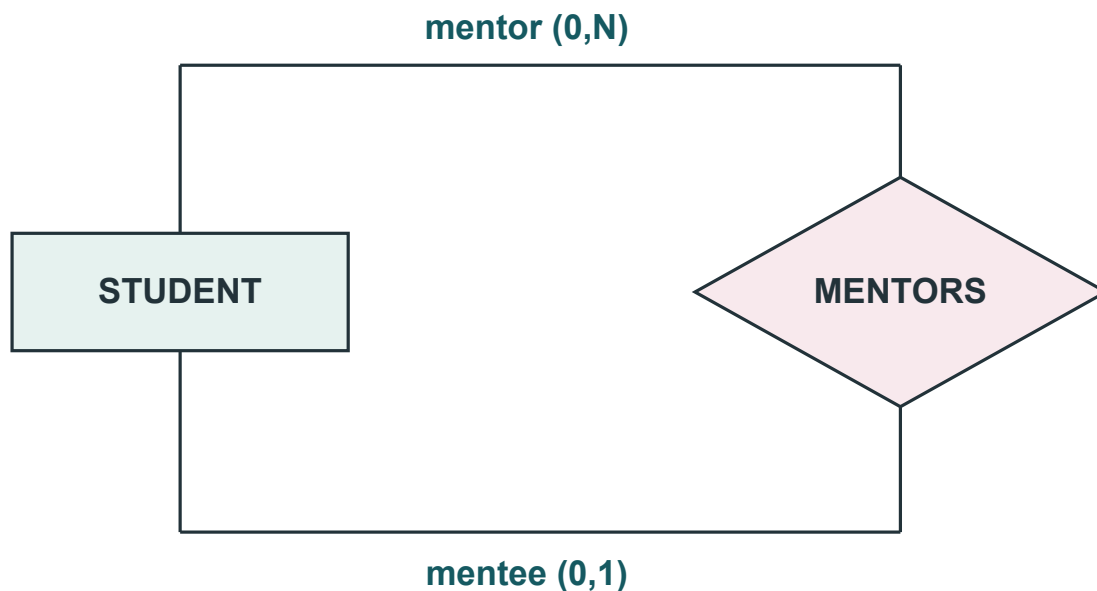
Read beside	Meaning	Example
INSTRUCTOR (0,N)	Zero or more TEACHES instances	I1 has 2; I3 may have 0.
SECTION (1,1)	Exactly one TEACHES instance	DB101 / 1 has 1 instructor.

1. **Minimum first, maximum second.**
2. (1,1) beside SECTION does not mean one student per section.
3. The pair supplies both bounds; a double participation line is unnecessary here.

Remember: Do not copy label positions from maximum-ratio notation.

Recursive Relationships: Mentoring

Students guide other students in a mentoring program. The guide is the mentor; the student receiving help is the mentee.



These are (min,max) pairs, not M:N ratios. Count one student's participations in the named role, not the opposite side.

Role	Ask about one student	Allowed
mentor (0,N)	How many mentees can this student guide?	0 or more
mentee (0,1)	How many mentors can this student have?	0 or 1

1. S101 guides S102 and S103: S101 participates twice as mentor; S102 and S103 each participate once as mentee.

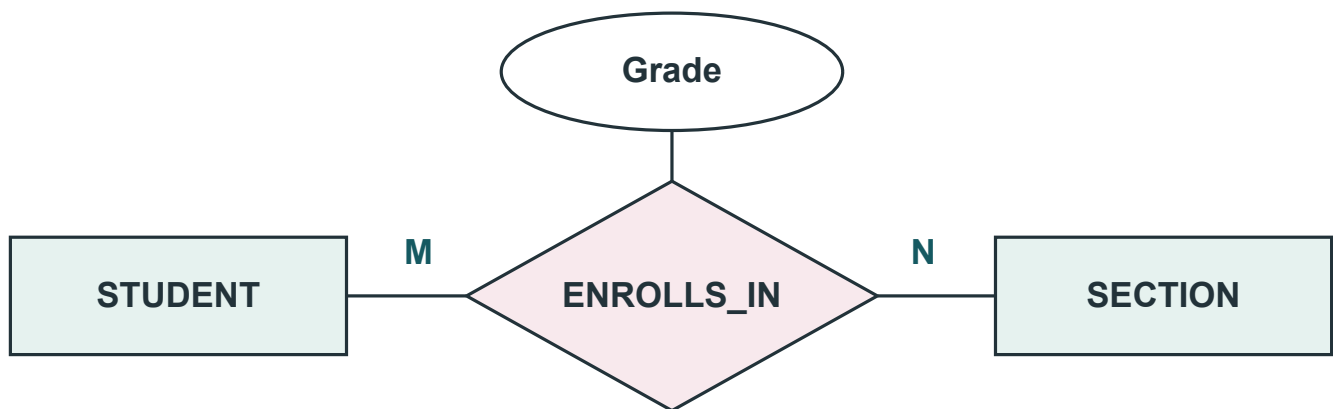
2. Reversing mentor and mentee changes the fact. These limits alone do not forbid self-mentoring or cycles.

Remember: One mentor may guide many mentees; each mentee may have at most one mentor.

Relationship Attributes: Grade Belongs to Enrollment

The college records a grade for each student's enrollment. S101 earns 80 in DB101 / 1 and 90 in CS102 / 1.

Student	DB101 / 1	CS102 / 1
S101	80	90
S102	70	No enrollment



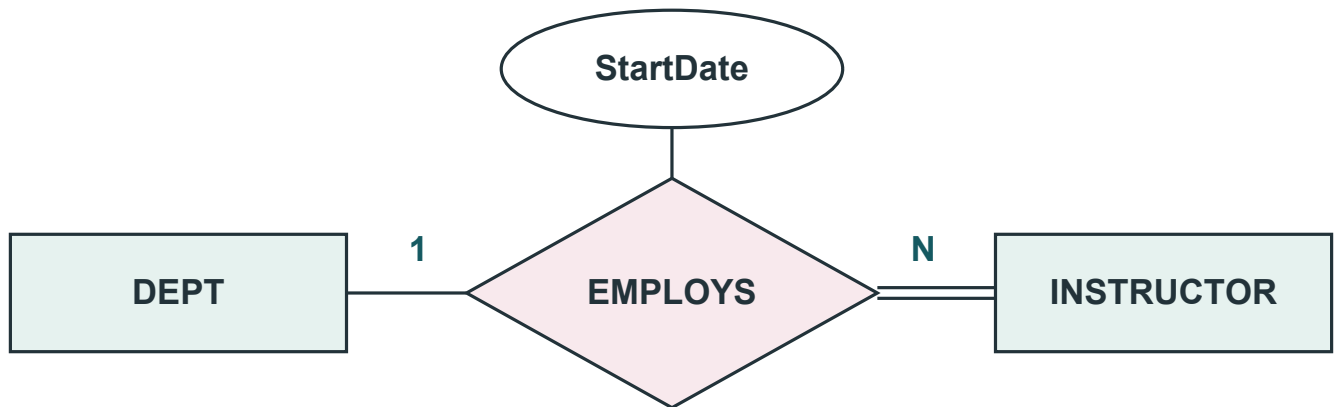
ER relationship close-up: Grade attaches to ENROLLS_IN. Identification is omitted.

1. S101 has **80 and 90**. One student-level grade cannot describe both.
2. DB101 / 1 has **80 and 70**. One section-level grade cannot describe both.
3. No enrollment, an enrollment with an unknown grade, and a recorded zero are different.

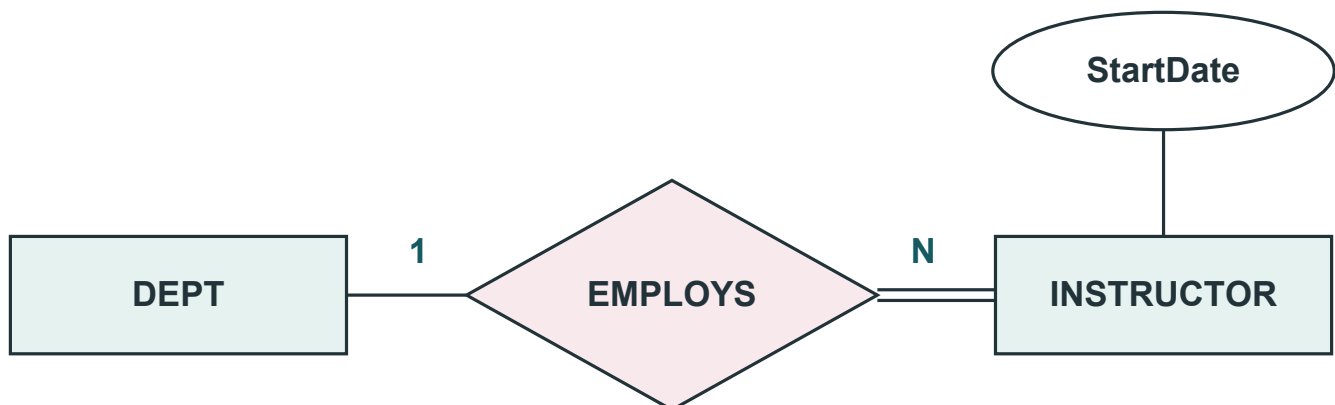
Remember: Grade belongs to this M:N enrollment, not to STUDENT or SECTION alone.

Relationship Attributes: StartDate in 1:N

Track when instructors join departments such as D1. Each instructor has one current department.



A: StartDate describes EMPLOY. Elapsed time can be calculated using an as-of date. Keys omitted.



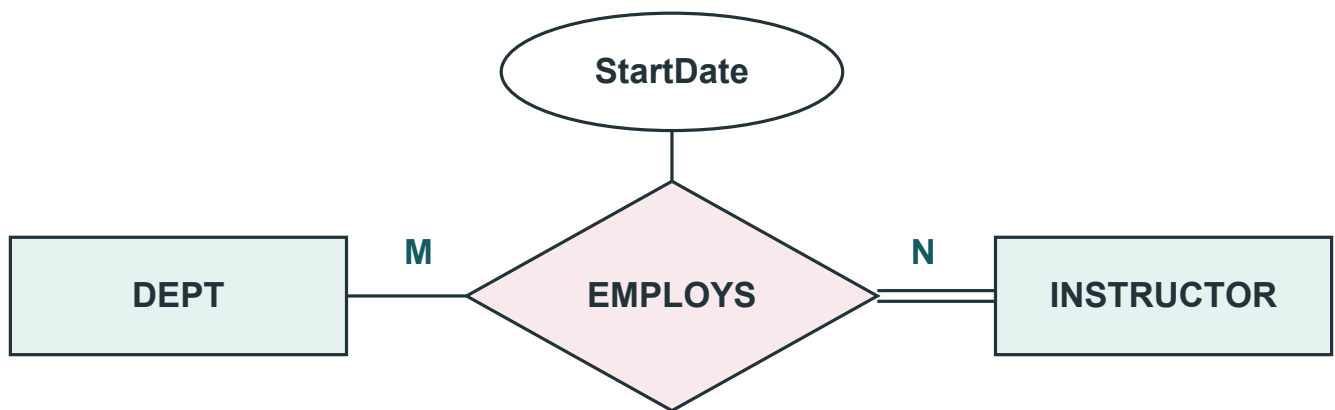
B: Date on INSTRUCTOR. Choose A or B, not both. Keys omitted.

Instructor	Current department	StartDate
I1	D1	2025-08-01
I2	D1	2026-02-01

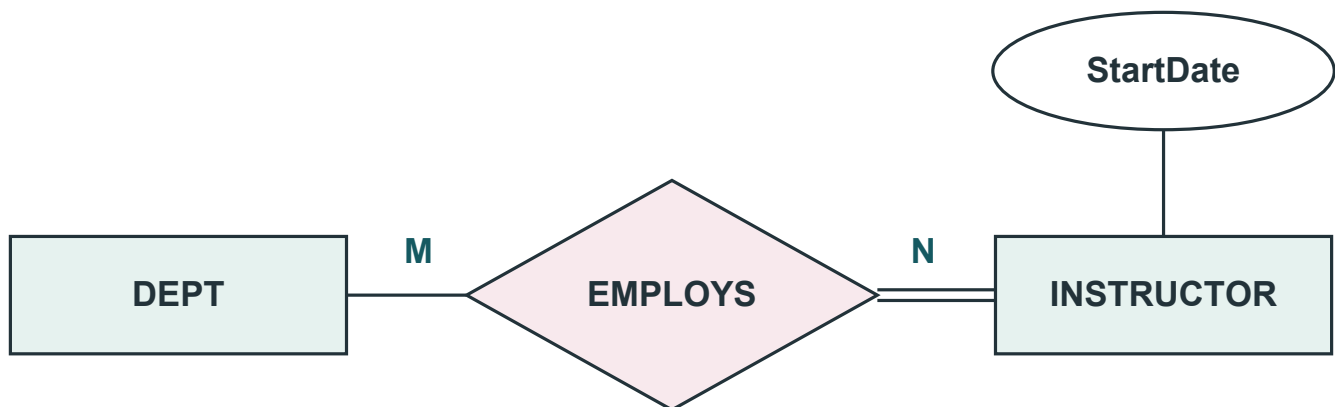
Remember: Both work: each instructor has one department. DEPT cannot store their different start dates.

Relationship Attributes: StartDate in M:N

New rule: allow multiple departments. I1 now works in D1 and D2.



A: Correct. Each department-instructor pair has its own StartDate. Keys omitted.



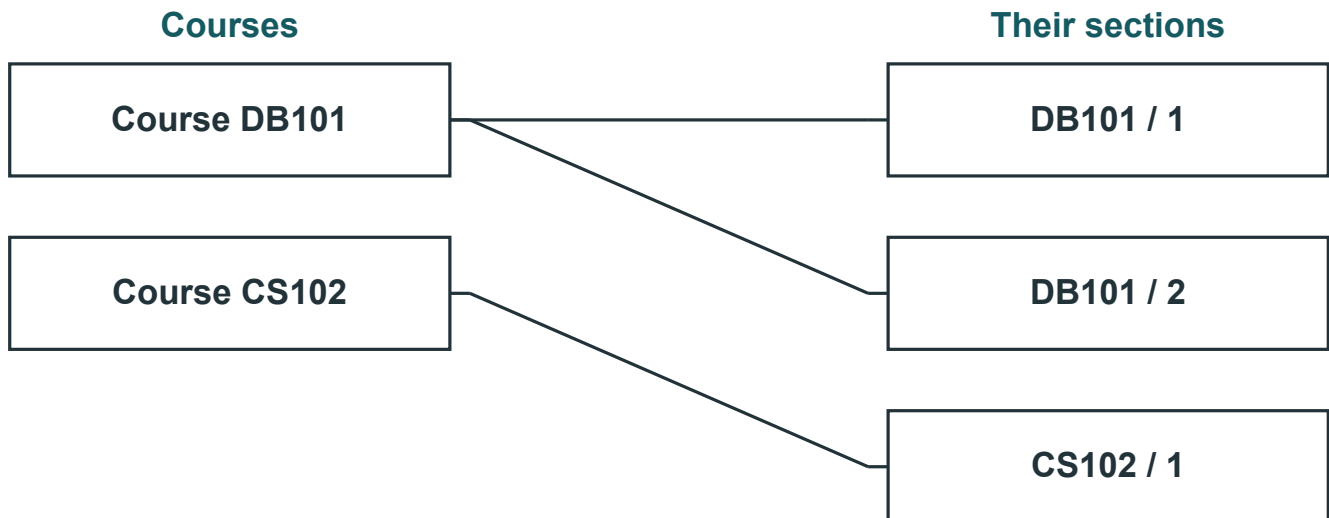
B: Incorrect for these facts. I1 needs two dates, not one.

Instructor	Current department	StartDate
I1	D1	2025-08-01
I2	D1	2026-02-01
I1	D2	2026-09-01

Remember: I1 has two dates; D1 also has two. StartDate needs the department-instructor pair.

Weak Entities: COURSE and SECTION

One course can have several classes, called sections. In this example, section numbers restart for each course; no global section ID exists.



Fall 2026 only: two courses, three sections. DB101 / 1 means section 1 of course DB101.

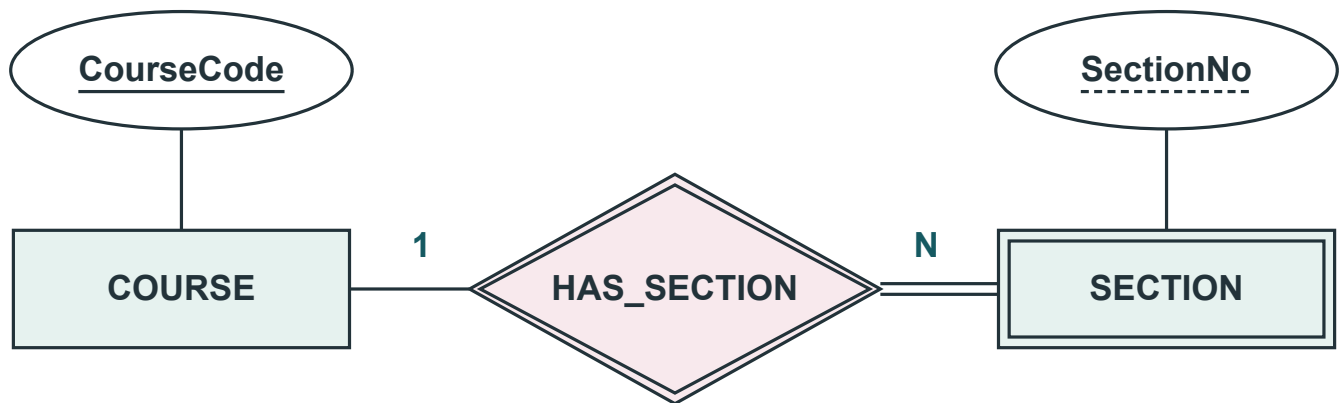
CourseCode	SectionNo	Which class?
DB101	1	DB101's first section
DB101	2	DB101's second section
CS102	1	CS102's first section

These records are valid. CourseCode is unique across courses; SectionNo is unique within one course. SECTION has no separate unique ID.

Remember: SECTION is weak here: identifying a class needs its course plus its section number.

Weak Entities: Owner and Partial Key

COURSE is the owner. SectionNo is the partial key: it distinguishes sections only within the same course.



Each section has one course. CourseCode stays on COURSE, reached through HAS_SECTION.

CourseCode	SectionNo	Identified section
DB101	1	DB101 / 1
DB101	2	DB101 / 2
CS102	1	CS102 / 1

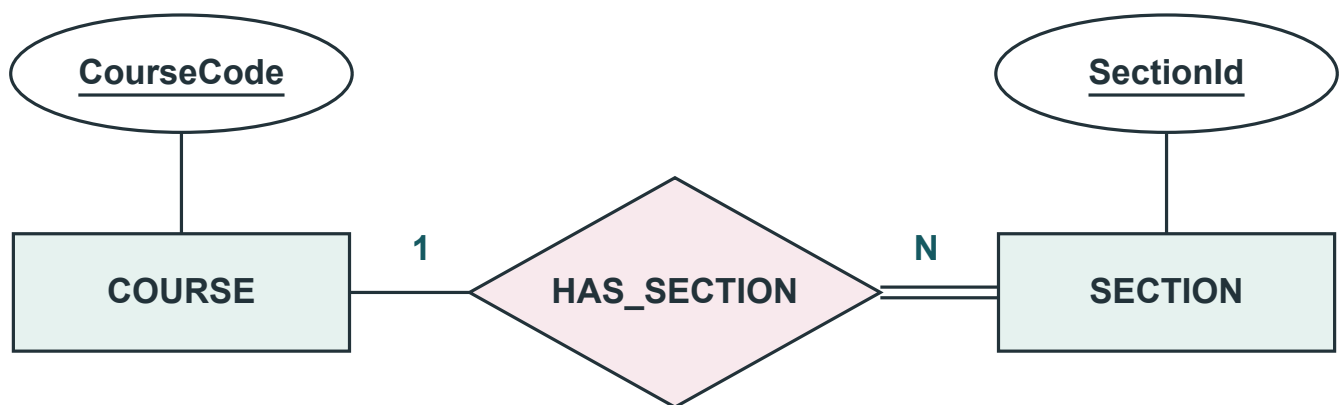
1. **Double rectangle:** SECTION needs its owner to identify it.
2. **Dashed underline:** 1 repeats across courses; SectionNo is partial.
3. **Double diamond:** HAS_SECTION supplies DB101 for DB101 / 1.
4. **Double connecting line:** a section with no course is invalid.

Remember: Owner CourseCode + partial key SectionNo = one section. A section cannot exist without its course.

Strong Entities: A Separate SectionId

Keep the same three classes. New rule: give each a globally unique SectionId. Q01 now identifies DB101 / 1 by itself.

SectionId	Course	SectionNo
Q01	DB101	1
Q02	DB101	2
Q03	CS102	1



Changed variant: SectionId replaces owner-based identification. SectionNo is omitted, not removed.

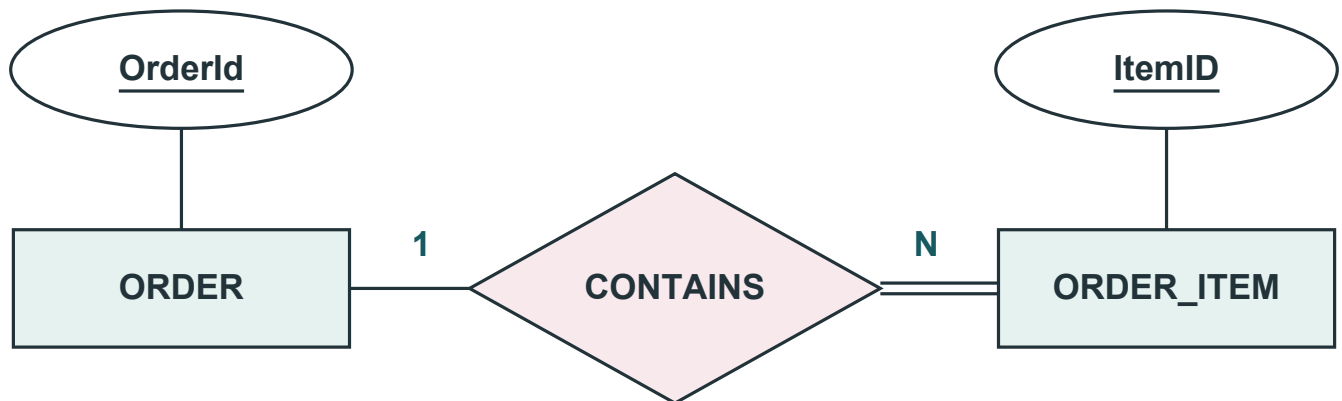
- 1. Before:** DB101 + 1 identifies the class. **Now:** Q01 alone does; SECTION is regular (strong).
- 2.** The rectangle and diamond become single; SectionId has a solid underline.
- 3.** The connecting line stays double: every section still needs a course.

Remember: Needing a course is not the same as needing its identity for identification.

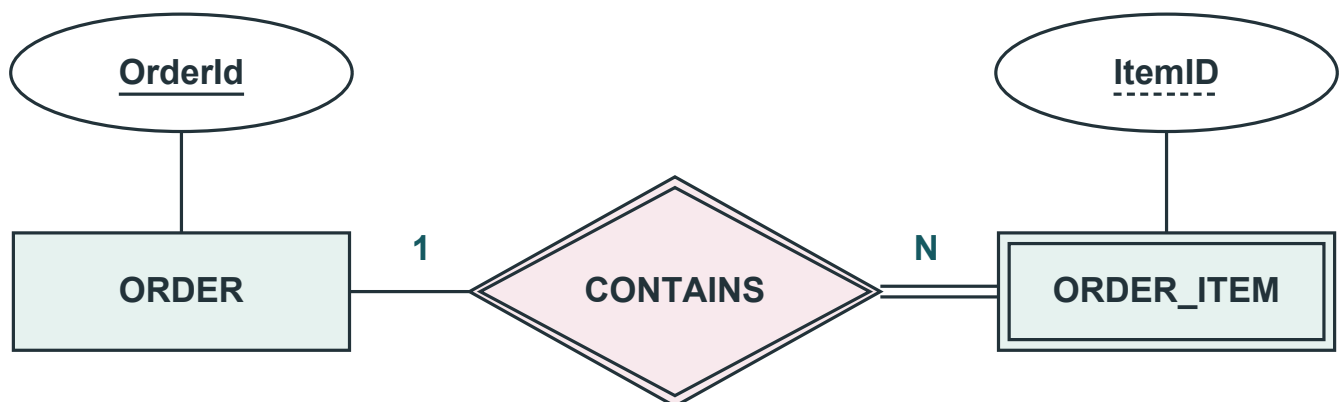
ORDER_ITEM: Wrong and Repaired

ItemID restarts within each order. O10 and O20 are shop order IDs.

OrderID	ItemID	Product	Quantity
O10	1	Notebook	2
O10	2	Pen	3
O20	1	Notebook	2



A: Wrong. Which two rows violate the solid ItemID underline?



B: Repaired. ItemID is unique within one ORDER: use a dashed underline.

Remember: ItemID 1 matches O10 / 1 and O20 / 1. ItemID is a partial key; we need OrderID + ItemID.

ORDER_ITEM: Compare the Same Item Records

The same three shop items appear twice below. Product and Quantity describe an item; they do not identify it.

Wrong: ORDER_ITEM without its owner context

ItemID	Product	Quantity
1	Notebook	2
2	Pen	3
1	Notebook	2

A solid underline on ItemID claims global uniqueness. Two distinct items have ItemID 1, even with identical descriptions.

Repaired: ORDER_ITEM with its owner reference

Owner OrderId	ItemID	Product	Quantity
O10	1	Notebook	2
O10	2	Pen	3
O20	1	Notebook	2

O10 / 1 selects one item. OrderId comes through CONTAINS; it need not become an attribute oval on ORDER_ITEM.

Remember: Wrong is the complete-key claim. Repaired identity is OrderId + ItemID; ItemID stays order-local.

Library: Background and Requirements

Original worked case: a library records individual borrowing events.

Rule	What must be represented
R1	Each BOOK represents an edition identified by ISBN. It may have no copies or many copies.
R2	Every COPY belongs to one BOOK. CopyNo is unique only within that book; there is no global copy ID.
R3	Each MEMBER has a unique MemberId and may make no loans or many loans.
R4	Each LOAN has a unique LoanId and one BorrowedOn date. It belongs to exactly one member and concerns exactly one copy.
R5	A copy may appear in no loans or many loans over time. A member may borrow that same copy again.

Draw these required attributes and relationships. Loan limits, fines and returns are outside this example.

Remember: Mark objects, identifiers and rules before choosing symbols.

Library: Books, Copies and Loans

B1 and B2 stand for different book ISBNs. M1 is a member. The library records copies and each borrowing event.

Book owner (ISBN alias)	CopyNo	Identified COPY
B1	1	B1 / 1
B2	1	B2 / 1

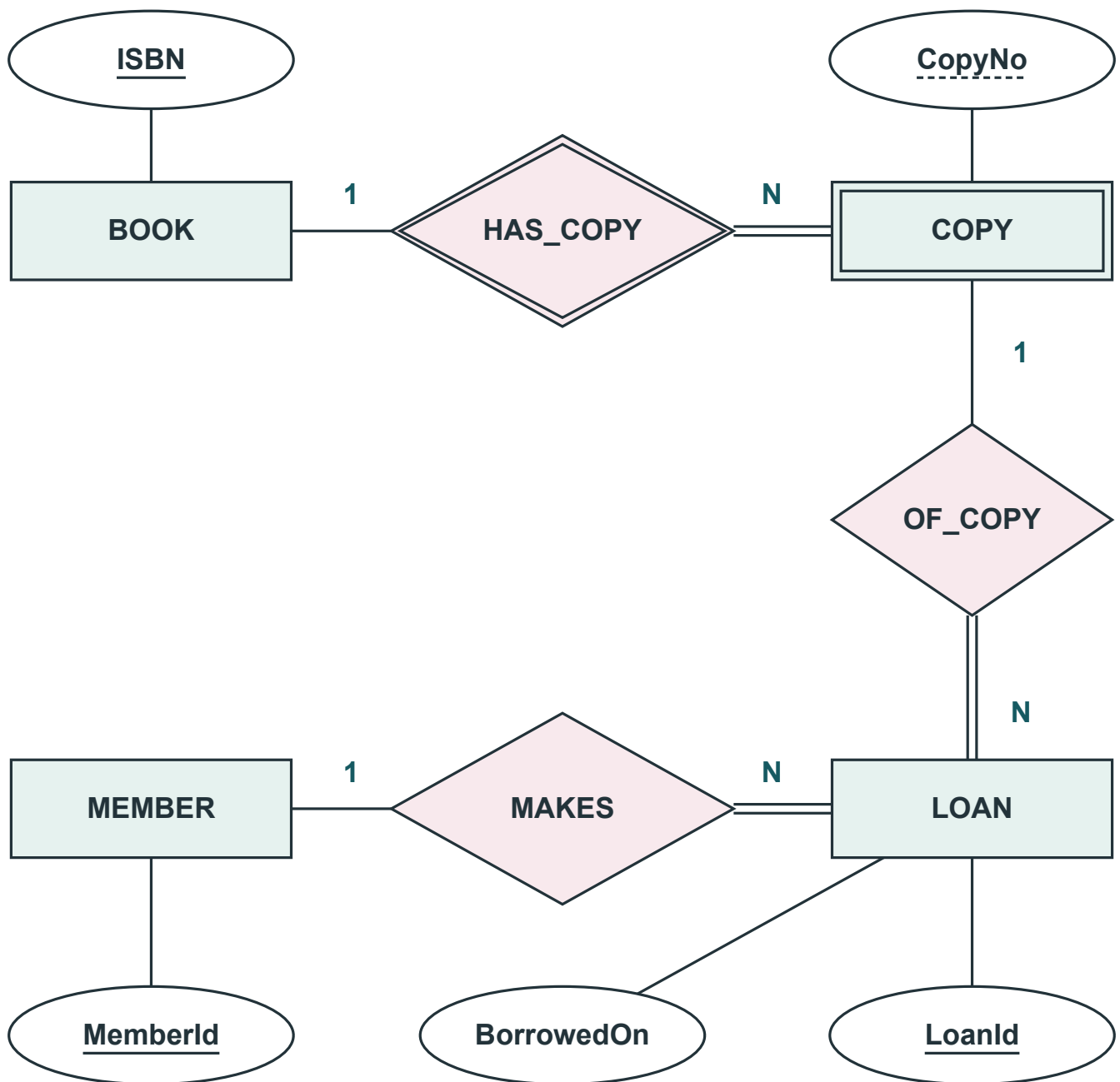
LoanId	Member	Copy	BorrowedOn
L1	M1	B1 / 1	2026-09-01
L2	M1	B1 / 1	2026-09-08

- 1. COPY:** two copies have CopyNo 1. Their book owners distinguish them.
- 2. LOAN:** L1 and L2 are two events, not two members or two copies.
- 3. BorrowedOn:** each loan has one date. M1 does not have one date for all loans.

Remember: COPY needs ISBN + CopyNo. LOAN has its own LoanId, even when the same member borrows the same copy again.

Library: The Complete ERD

M1 borrows B1 / 1 as L1. Where do you follow the owner and find the date?

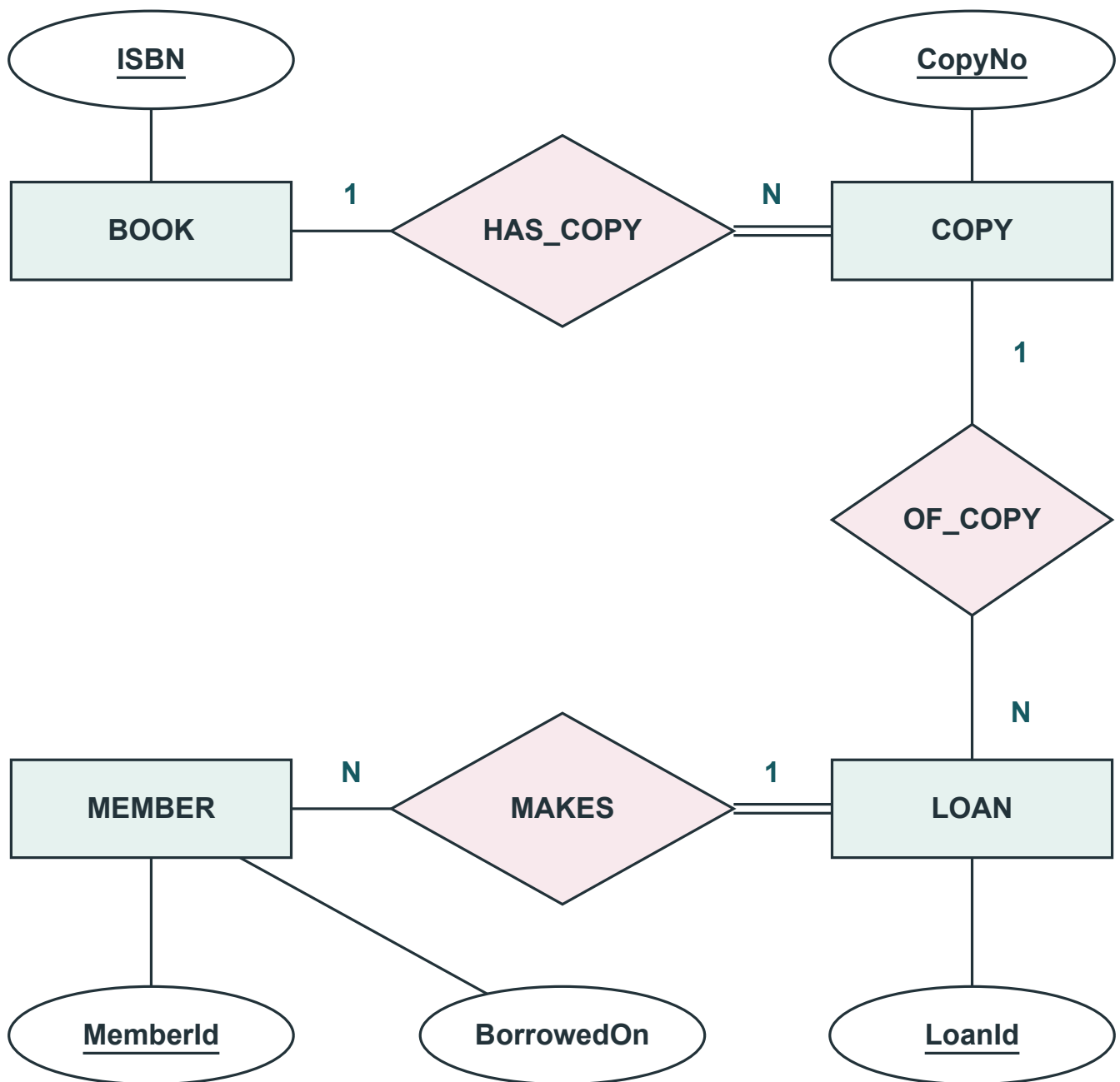


Example: M1 makes L1 for copy B1 / 1 on 2026-09-01. Trace MEMBER, LOAN, COPY, then BOOK.

Remember: Trace each requirement to one visible key, shape, attribute or connection.

Library: The Incorrect ERD

INCORRECT example. Keep R1-R5 fixed when finding the four errors.



Test B1 / 1 and B2 / 1; M1's loans L1 (Sep 1) and L2 (Sep 8). R1-R5 permit these facts.

Remember: For each error, name the violated requirement and a concrete counterexample.

Library: Four Errors and Their Repairs

Keep R1-R5 fixed. Do not change the story to make the wrong diagram work.

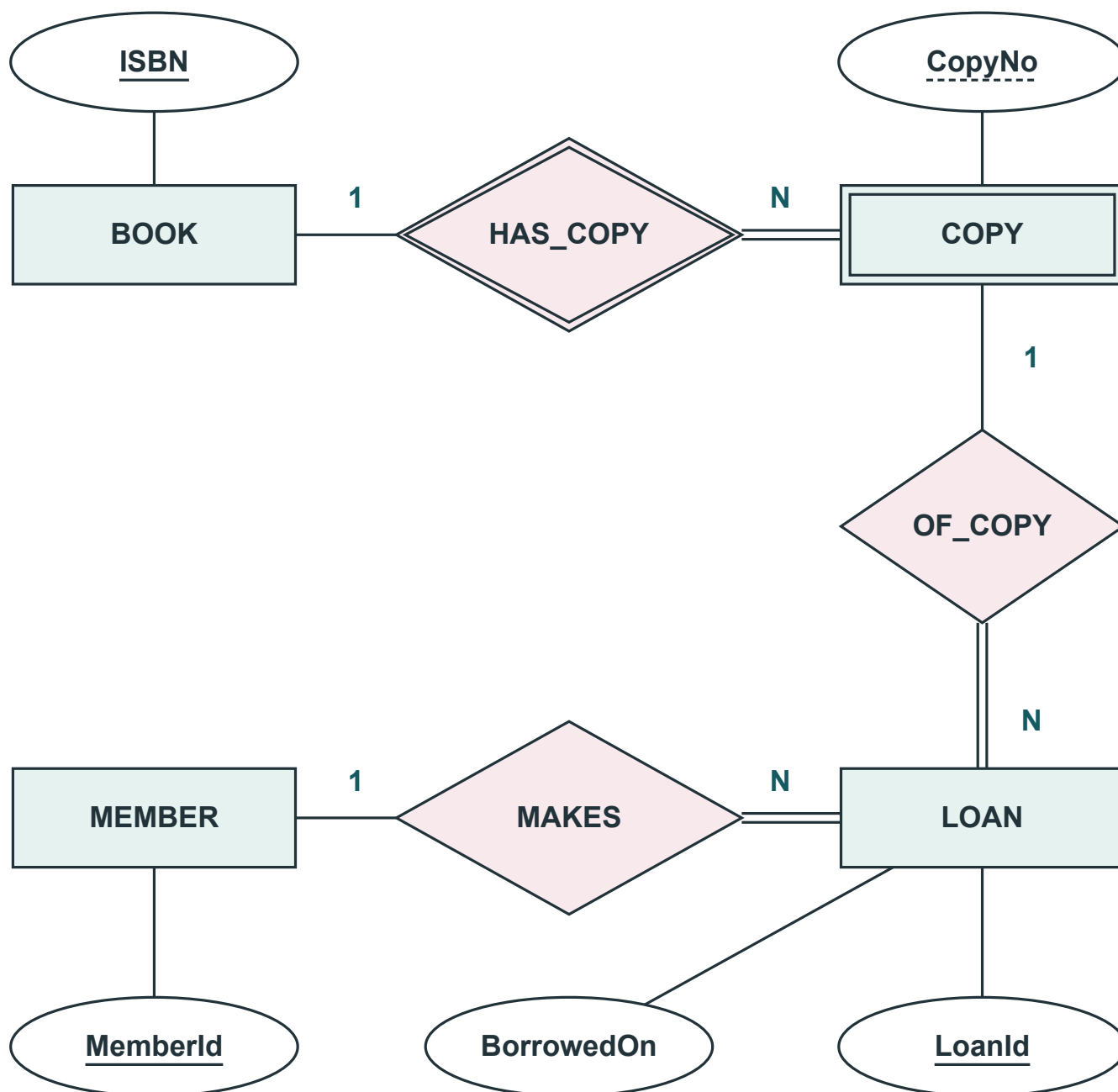
Wrong part	Counterexample	Required repair
COPY is regular; CopyNo is a full key	B1/1 and B2/1 are both allowed.	Weak COPY; partial CopyNo; identifying HAS_COPY.
MAKES ratio is N:1	One member makes L1 and L2.	1 near MEMBER; N near LOAN.
LOAN participation in OF_COPY is partial	A loan with no copy is forbidden by R4.	Double line at LOAN.
BorrowedOn is on MEMBER	M1 borrows on Sep 1 and Sep 8.	Move the date to LOAN.

BorrowedOn moving here is necessary because one member can make many loans with different dates.

Remember: A correction needs a reason, not just a differently drawn symbol.

Library: The Corrected ERD

Can B1 / 1 and B2 / 1 now coexist? Which repaired symbols permit this?



Now B1 / 1 and B2 / 1 have different owners; M1's L1 and L2 each keep their own date and copy.

Remember: Compare the repaired parts against the counterexamples, one at a time.

Ternary Relationships: Course Approval

Next, one relationship connects three entities: a student asks an instructor for permission to take a course.

Background	What we record
Request	Alex wants permission to take Database Systems.
Decision	Instructor Morgan approves Alex for that course.
Question later	Who approved Alex for Database Systems?

1. Scope: record granted permissions for Fall 2026. This is an invented teaching example, not university policy.

2. Approval is not enrollment: permission does not say that the student joined a section or received a grade.

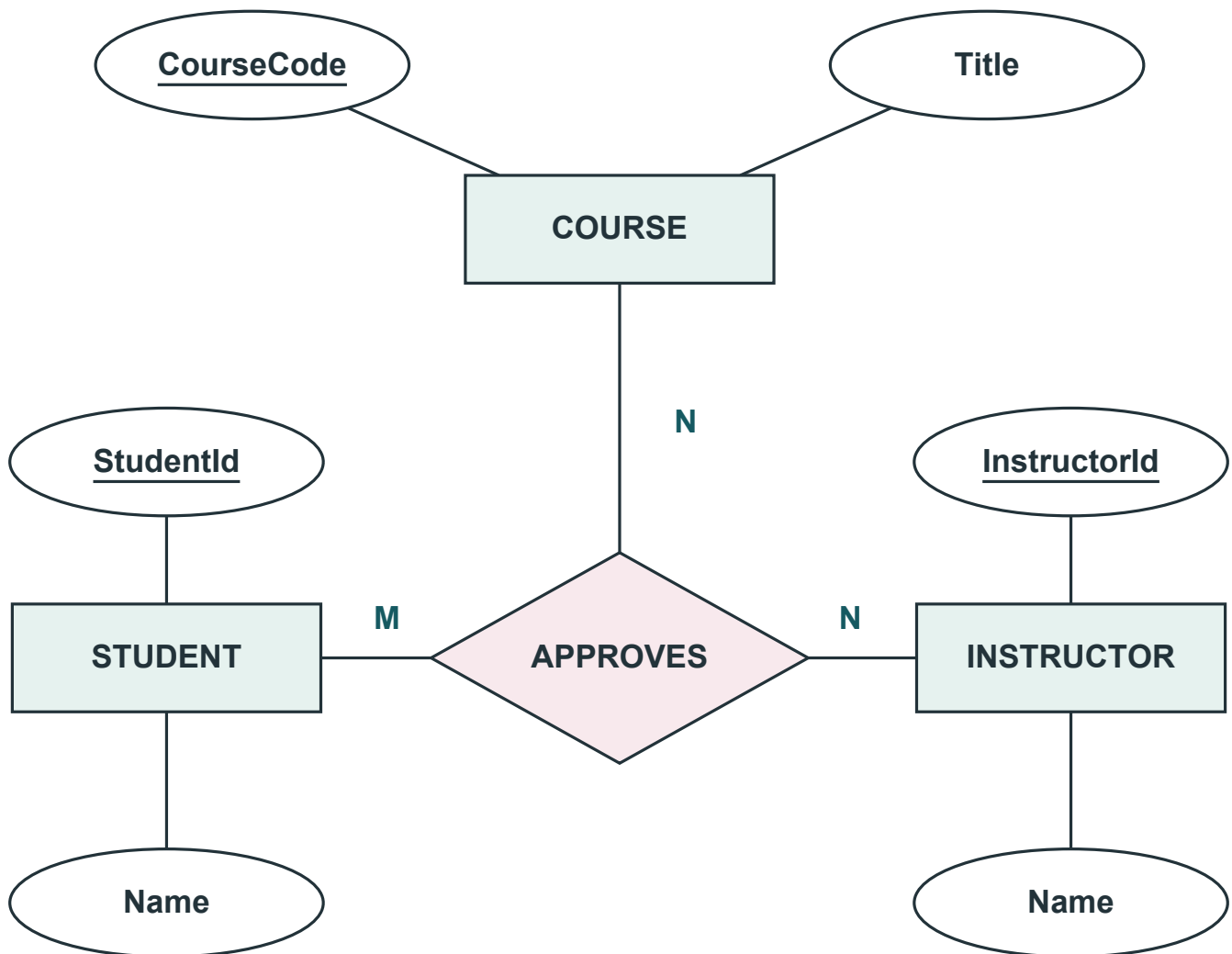
3. Starting rules: any participant may have no approvals or many. Record each student-instructor-course combination once.

First see the complete model and its records. Then change one rule at a time.

Remember: APPROVES means: this instructor approved this student for this course.

APPROVES: Student, Instructor and Course

Continue the permission example: one instructor approves one student for one course. All three belong in the same fact.



Complete for this example: identifiers and descriptive attributes are shown. Sections, enrollment and grades are outside its scope.

Underlines: each entity's own key. **Single connecting lines:** participation is optional. M/N labels impose no fixed upper bound; they are not current counts.

Remember: One **APPROVES** fact needs a student, an instructor and a course together.

APPROVES: The Participant Records

Use these same identifiers in every approval record.

STUDENT

StudentId	Name
S101	Alex Lin
S102	Blair Wu
S103	Casey Chen

INSTRUCTOR

InstructorId	Name
I1	Morgan
I2	Taylor

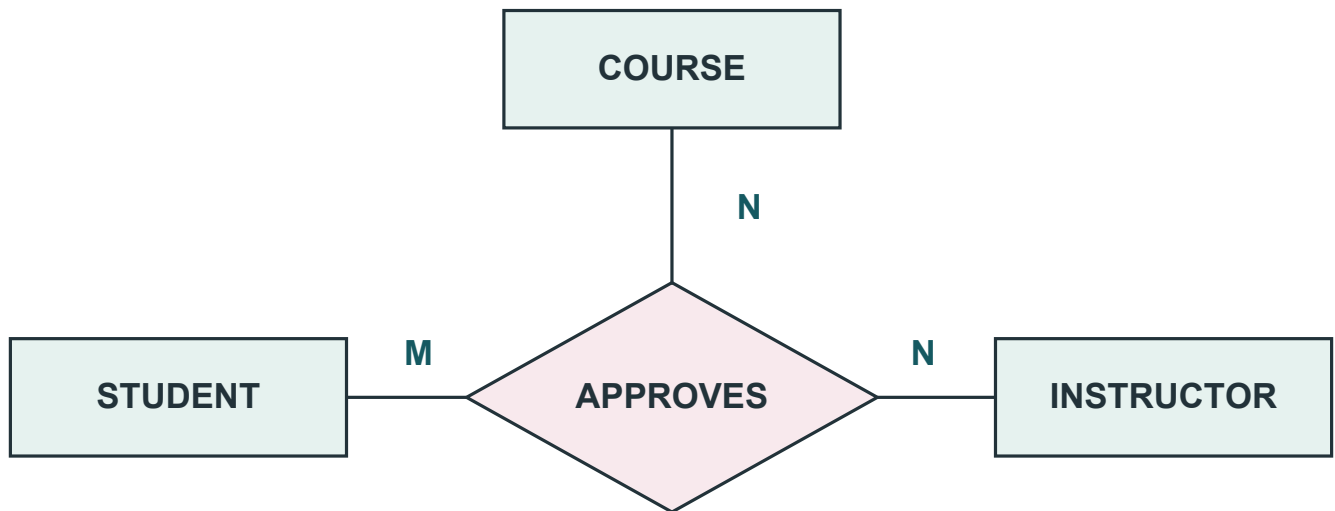
COURSE

CourseCode	Title
DB101	Database Systems
CS102	Programming

Remember: S103 exists with no approval. These are entity records, not approval facts or a SQL design.

APPROVES: Read One Complete Record

Who approved Alex, and for which course? Read one whole row.



Close-up of the preceding complete ERD; attributes omitted, rules unchanged.

StudentId	InstructorId	CourseCode
S101	I1	DB101
S101	I2	CS102
S102	I1	CS102

- 1. First row:** Morgan approved Alex for Database Systems.
- 2. Second row:** Taylor, not Morgan, approved Alex for Programming.
- 3. Third row:** Morgan approved Blair for Programming.
- 4.** Does Morgan approve Alex for Programming here? No: S101/I1/CS102 is not recorded. The next page tests a tempting wrong inference.

Remember: Read all three participants in the same row: who approved which student for which course.

APPROVES: Separate Pairs Do Not Prove Approval

Did Morgan (I1) approve Alex (S101) for Programming (CS102)? Check the same three approvals, one row at a time.

Student	Instructor	Course	Why not the requested approval?
S101	I1	DB101	Different COURSE
S101	I2	CS102	Different INSTRUCTOR
S102	I1	CS102	Different STUDENT

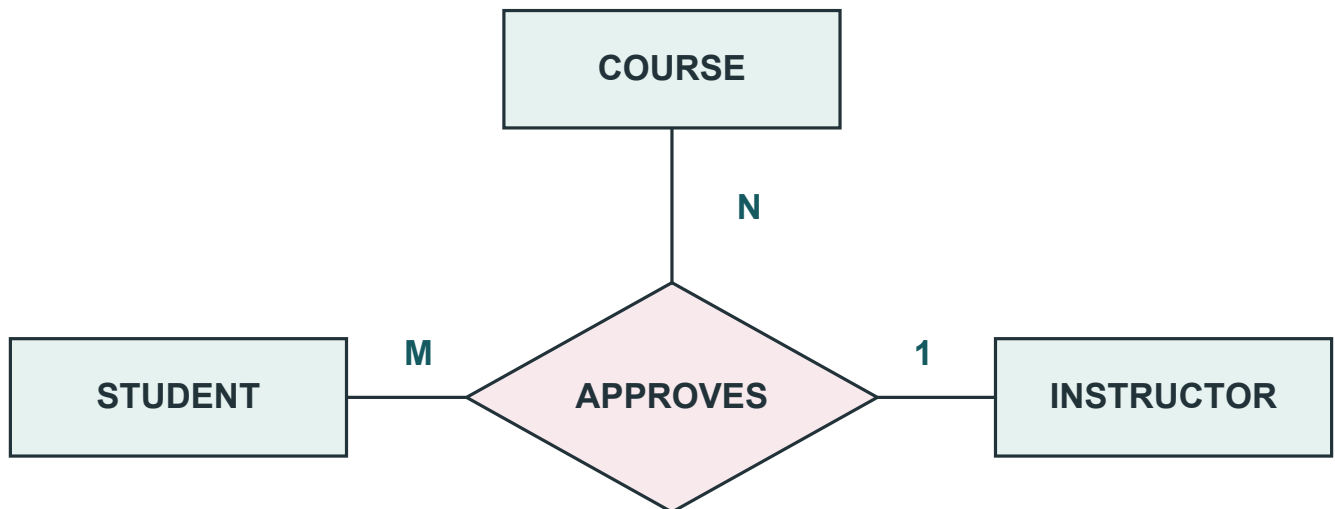
The first row connects Alex to Morgan, the second Alex to Programming, and the third Morgan to Programming. **No single row connects all three.**

A new S101 / I1 / CS102 row would record a NEW approval. Those separate connections do not prove it already exists.

Remember: No matching approval is recorded. Do not combine parts of different rows; this does not mean the request was rejected.

Ternary Cardinality: Fix Two Participants

Keep the same three approvals. New rule for this variant: at most one instructor may approve a particular student for a particular course.



The 1 is beside INSTRUCTOR; first fix STUDENT and COURSE.

Check against the original records	Result
I1 appears with S101/DB101 and S102/CS102	Allowed: different student-course pairs.
Add S101, I2, DB101	Rejected: S101/DB101 already has I1.

1. Under this rule, the student-course pair identifies an approval instance.
2. For n roles, fix the other n-1 participants before interpreting a 1.

Remember: The 1 limits who completes a fixed pair, not the instructor's total approvals.

UNIVERSITY: Background and Scope

Separate textbook case: UNIVERSITY stores organization and grades across years. Its records and rules replace the earlier campus example.

Part of the school	What the database records
College and departments	One college C-A contains departments D1 and D2.
People and roles	Three instructors; five students; dean and chair appointments.
Teaching	Courses, their term-specific sections and their teachers.
Enrollment	Which student takes which section, with a grade.

A COURSE describes a subject, such as DB101. A SECTION is one offering in a particular term. Each section needs one teacher and at least five students in this example.

Next: requirements, an incorrect ERD, sample records, and six repairs leading to Figure 3.20. Tables illustrate ER facts; SQL design comes later.

Remember: Here SECTION has a global SecId: it is strong, unlike the earlier one-term weak SECTION example.

UNIVERSITY: Organization Requirements

A college groups departments. Its dean leads the college; a chair leads one department. Figure 3.20 supplies the following rules.

Rule	What must be represented
R1	A college may administer many departments or none. Each department belongs to exactly one college.
R2	Each college has exactly one dean. An instructor may be dean of at most one college, or none.
R3	Each department has exactly one chair. An instructor may chair at most one department, or none. Record CStartDate for the appointment.
R4	Each instructor works for one department. A department may employ many instructors or none.

CName identifies COLLEGE. DCode and DName each independently identify DEPT. Id identifies INSTRUCTOR.

Remember: Translate each rule in both directions. A department needs a chair; an instructor need not be a chair.

UNIVERSITY: Teaching and Enrollment Requirements

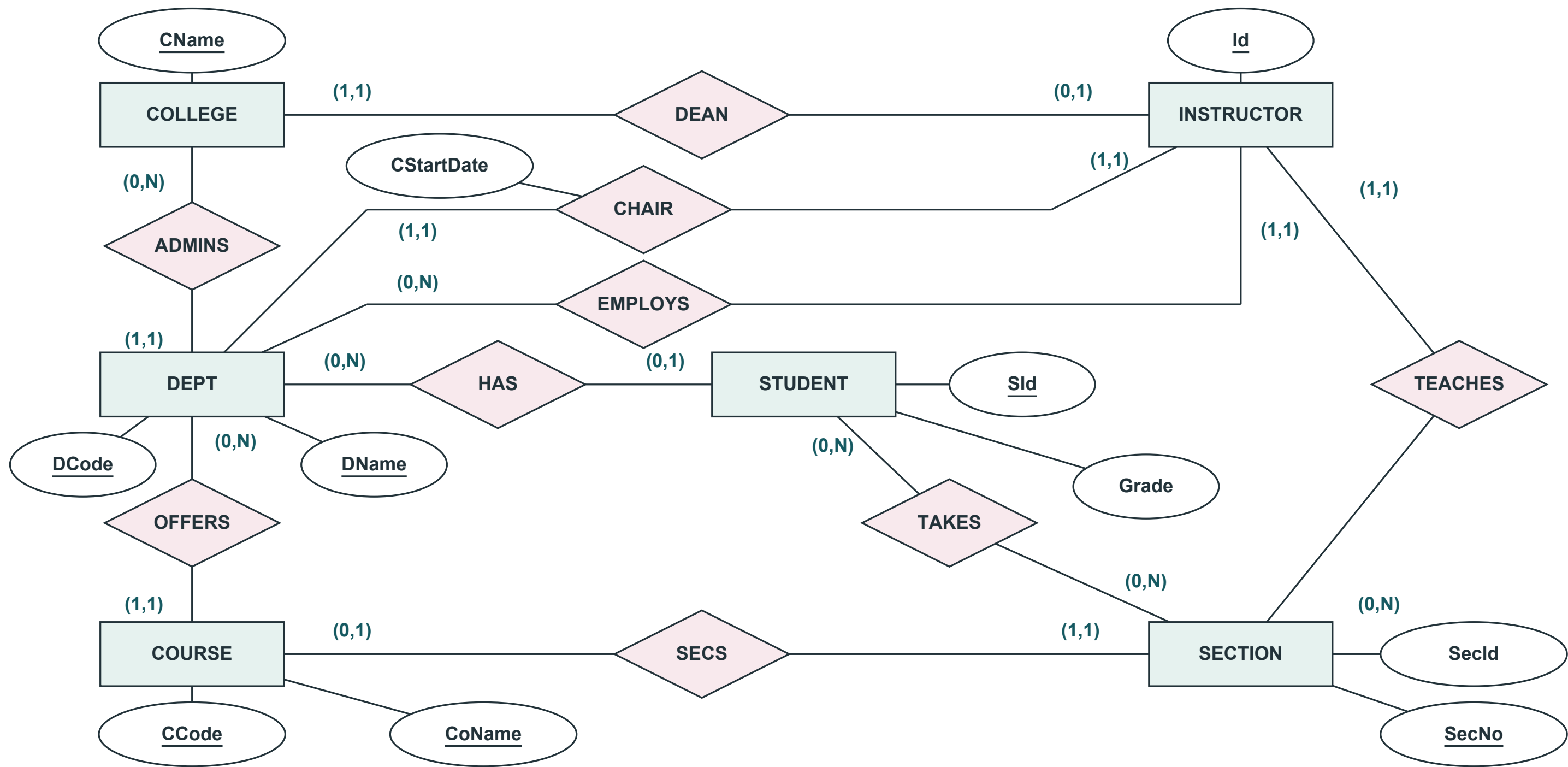
Keep the course, its term-specific offering, and the student separate.

Rule	What must be represented
R5	A student has zero or one department; a department may have many students or none.
R6	Each course belongs to one department. A department may offer many courses or none.
R7	SecId identifies SECTION across all terms. Each section belongs to one course; a course may have many sections or none.
R8	Each section has exactly one instructor. An instructor may teach many sections or none.
R9	A student may take many sections or none. Each section needs at least five students. Record Grade for each student-section pair when available.

Remember: R5 follows the figure's optional department assignment. The prose requires one; this exercise uses the figure.

UNIVERSITY: The Incorrect ERD

INCORRECT teaching draft. Compare it with R1-R9 and the sample records that follow.



Deliberately altered from Figure 3.20. This is NOT the answer. Non-key entity attributes other than SecNo are omitted; their values follow in the tables.

Remember: Check the SECTION key, TEACHES labels, Grade, the TAKES minimum, CHAIR participation and the SECS maximum.

UNIVERSITY Records: COLLEGE and DEPT

C-A is the sample college name; D1 and D2 are department codes.

COLLEGE

CName	COffice	CPhone
C-A	A100	02-0000-1000

DEPT

DCode	DName	DOffice	DPhone
D1	Computing	A101	02-0000-1100
D2	Information Systems	B101	02-0000-1200

All names, offices, phone numbers and later student details are fictional. The connections between these records come next.

Remember: CName identifies a college. DCode and DName each independently identify a department.

UNIVERSITY Records: COURSE and INSTRUCTOR

DB101 is a course. I1 is a person. Their connection needs a relationship.

COURSE

CCode	CoName	Level	Credits	CDesc
DB101	Database Systems	2	3	Tables and queries
CS102	Programming	1	3	Programs and data

INSTRUCTOR

Id	IName	IOffice	IPhone	Rank
I1	Morgan	A201	02-0000-2101	Professor
I2	Taylor	B201	02-0000-2102	Professor
I3	Jordan	A202	02-0000-2103	Lecturer

Level 1 = first year; Level 2 = second year. These rows do not yet say which section an instructor teaches.

Remember: Different course codes may not reuse CoName: both are keys in this textbook example.

UNIVERSITY Records: STUDENT

This separate UNIVERSITY dataset uses students U1-U5. FName, MName and LName mean first, middle and last name; Major is the study field.

SId	FName	MName	LName	Major
U1	Alex	-	Lin	CS
U2	Blair	-	Wu	Undeclared
U3	Casey	-	Chen	CS
U4	Drew	-	Huang	IS
U5	Evan	-	Liu	IS

A dash means no middle name recorded in this illustration. CS = Computing; IS = Information Systems. U2 has not declared a major.

Only attributes needed for this discussion are shown. U1-U5 are the same students used in the TAKES records.

Remember: SId identifies the student. SName contains FName, MName and LName.

UNIVERSITY Records: SECTION

Q101 and Q102 are DB101 offerings in different terms. Sem means semester; Bldg means building; DaysTime gives the meeting schedule.

SecId	SecNo	Sem	Year
Q101	1	Fall	2026
Q102	1	Spring	2027

SecId	Bldg	RoomNo	DaysTime
Q101	A	201	Thu 09:00-10:00
Q102	B	201	Thu 09:00-10:00

CRoom contains Bldg and RoomNo. Match the two display tables by SecId. SECS will connect both offerings to DB101.

Remember: SecNo 1 and RoomNo 201 repeat. SecId distinguishes the offerings; Bldg distinguishes the rooms.

UNIVERSITY Relationships: ADMINS, DEAN and CHAIR

Each table records one named relationship in the complete ERD.

ADMINS

CName	DCode
C-A	D1
C-A	D2

DEAN

CName	Id
C-A	I1

CHAIR

DCode	Id	CStartDate
D1	I1	2025-08-01
D2	I2	2026-02-01

Remember: CStartDate belongs to CHAIR: it records when the instructor became that department's chair.

UNIVERSITY Relationships: EMPLOYS and HAS

I3 works in D1. U2 has no HAS record. These are different relationships.

EMPLOYS

DCode	Id
D1	I1
D2	I2
D1	I3

HAS

DCode	SId
D1	U1
D1	U3
D2	U4
D2	U5

I3 appears in EMPLOYS but not CHAIR: employment does not require a chair appointment. U2 is present in STUDENT, not missing from the database.

Remember: This data follows Figure 3.20: a student may have no department. The prose instead requires one; do not mix the two versions.

UNIVERSITY Relationships: OFFERS, SECS and TEACHES

Trace Q101 back to DB101, D1 and instructor I1.

OFFERS

DCode	CCode
D1	DB101
D2	CS102

SECS

CCode	SecId
DB101	Q101
DB101	Q102

TEACHES

Id	SecId
I1	Q101
I1	Q102

Remember: D1 offers DB101; DB101 has Q101 and Q102; I1 teaches both offerings in different terms.

UNIVERSITY Relationships: TAKES and Grade

These are hypothetical completed records, not current student results.

TAKES: Q101

SId	SecId	Grade
U1	Q101	80
U2	Q101	85
U3	Q101	90
U4	Q101	75
U5	Q101	88

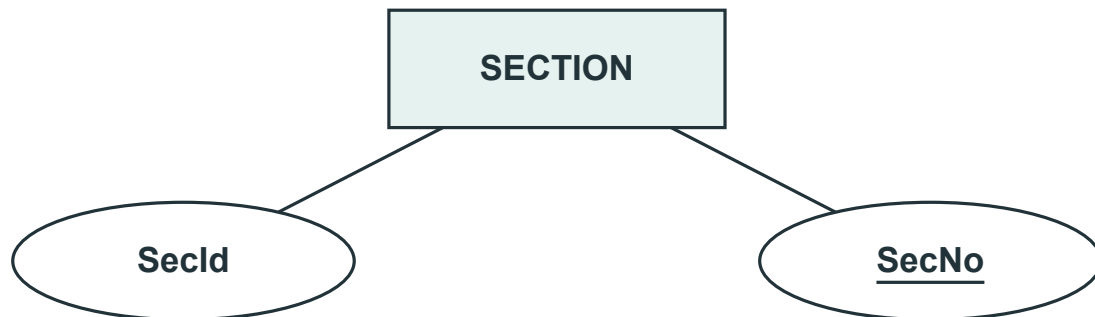
TAKES: Q102

SId	SecId	Grade
U1	Q102	90
U2	Q102	87
U3	Q102	92
U4	Q102	78
U5	Q102	91

Remember: Each section has five students. U1's 80 and 90 belong to different TAKES facts, not one student-level Grade.

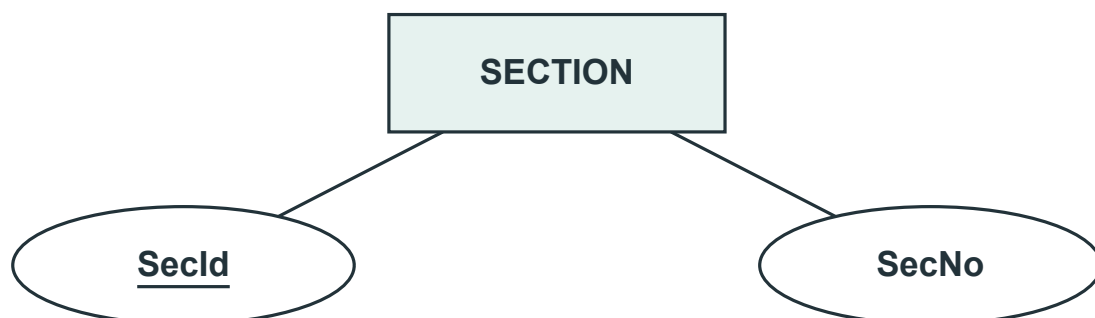
Repair 1: The SECTION Key

R7: SecId identifies a section across all terms. Can SecNo 1 identify just one?



INCORRECT: the solid underline claims that SecNo is unique.

SecId	SecNo	Sem	Year
Q101	1	Fall	2026
Q102	1	Spring	2027



CORRECT: Q101 and Q102 have different SecIds, although both have SecNo 1.

Remember: Remove the underline from SecNo. Underline SecId, as in Figure 3.20; SECTION remains a regular entity.

Repair 2: TEACHES Cardinality

R8: one teacher per section. Read each (min,max) beside its own entity.



INCORRECT: I1 would be limited to one section; a section could have no teacher or many.

Id	SecId
I1	Q101
I1	Q102

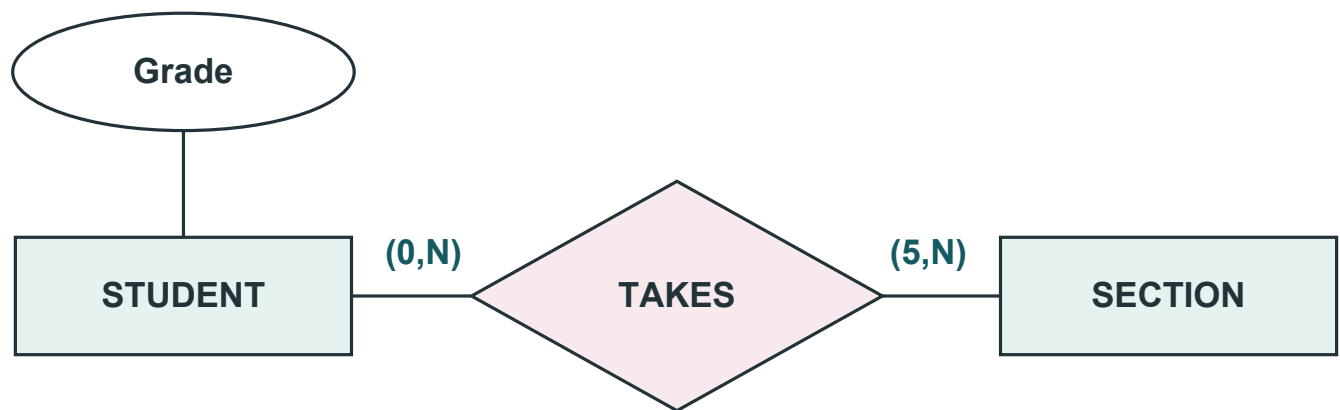


CORRECT: I1 can teach Q101 and Q102; each section must have exactly one teacher.

Remember: I1 teaches two sections. Put (0,N) beside INSTRUCTOR and (1,1) beside SECTION, as in Figure 3.20.

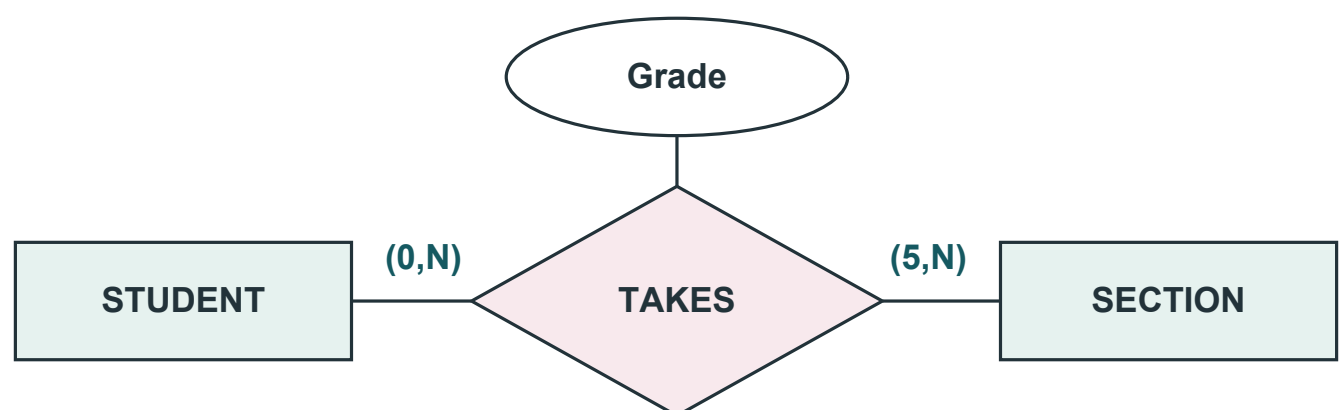
Repair 3: Grade Belongs to TAKES

R9: a grade describes a student in a particular section. Which connection needs the oval?



INCORRECT: one student-level Grade cannot identify the section for each result.

SId	SecId	Grade
U1	Q101	80
U1	Q102	90



CORRECT: Grade belongs to the student-section relationship, not either entity alone.

Remember: Move Grade from STUDENT to TAKES, as in Figure 3.20. Keep both results: U1/Q101 = 80 and U1/Q102 = 90.

Repair 4: Minimum Enrollment

R9: every section needs at least five students. Test a proposed removal from Q101.



INCORRECT: $(0,N)$ beside **SECTION** would permit even zero students. Grade and keys omitted.

Q101 students	Count	Meets R9?
U1, U2, U3, U4, U5	5	Yes
Proposed: remove U5/Q101	4	No

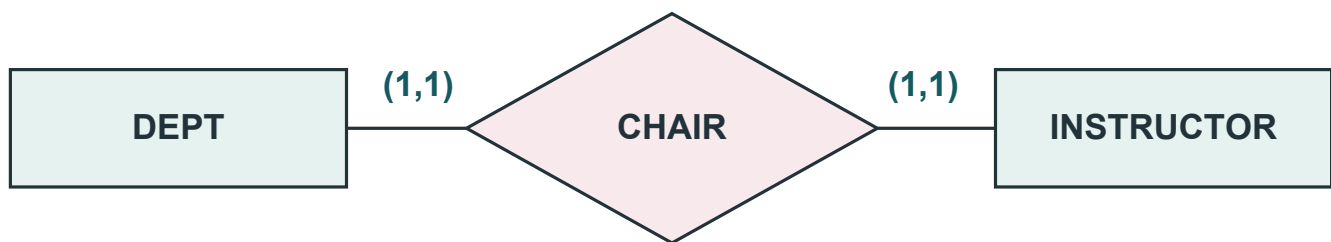


CORRECT: $(5,N)$ rejects the proposed four-student section. Keep the original five records.

Remember: Change only **SECTION**'s minimum from 0 to 5, as in Figure 3.20. **STUDENT** remains optional.

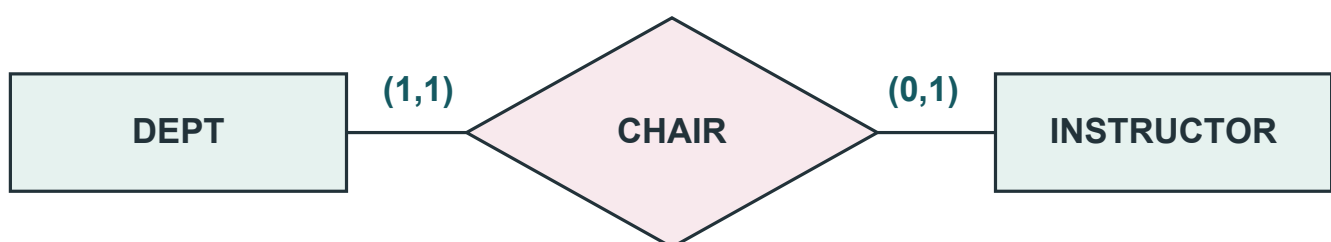
Repair 5: Optional CHAIR Participation

R3: a department needs a chair. Does that mean every instructor must chair a department?



INCORRECT: (1,1) beside INSTRUCTOR would require I3 to chair a department. Attributes omitted.

Instructor	Works in	Chairs
I1	D1	D1
I3	D1	None

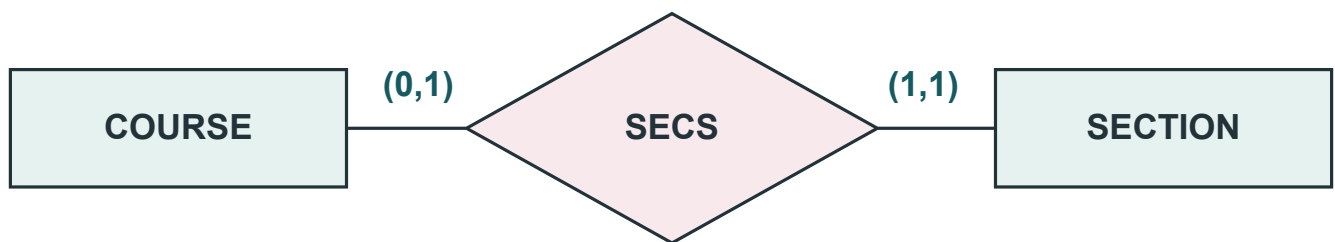


CORRECT: I3 can work in D1 without being a chair. Employment and chair appointments are separate.

Remember: No. Put (0,1) beside INSTRUCTOR, as in Figure 3.20. Keep DEPT (1,1): every department still needs a chair.

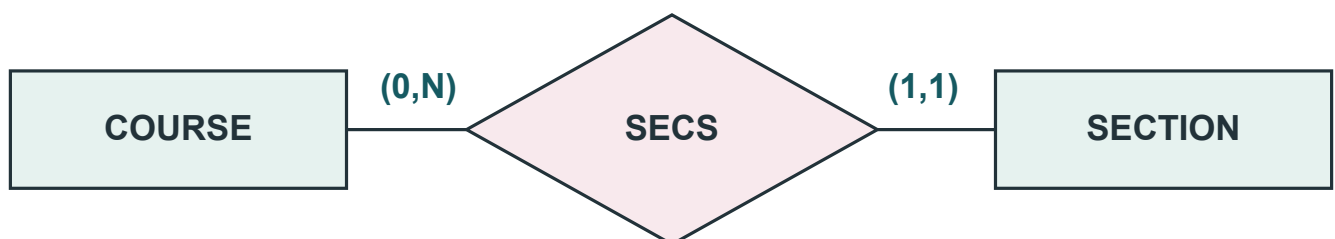
Repair 6: One Course, Many Sections

R7: store course offerings across different terms, not just one class per course.



INCORRECT: (0,1) beside COURSE would reject DB101's second section. Attributes omitted.

CCode	SecId
DB101	Q101
DB101	Q102

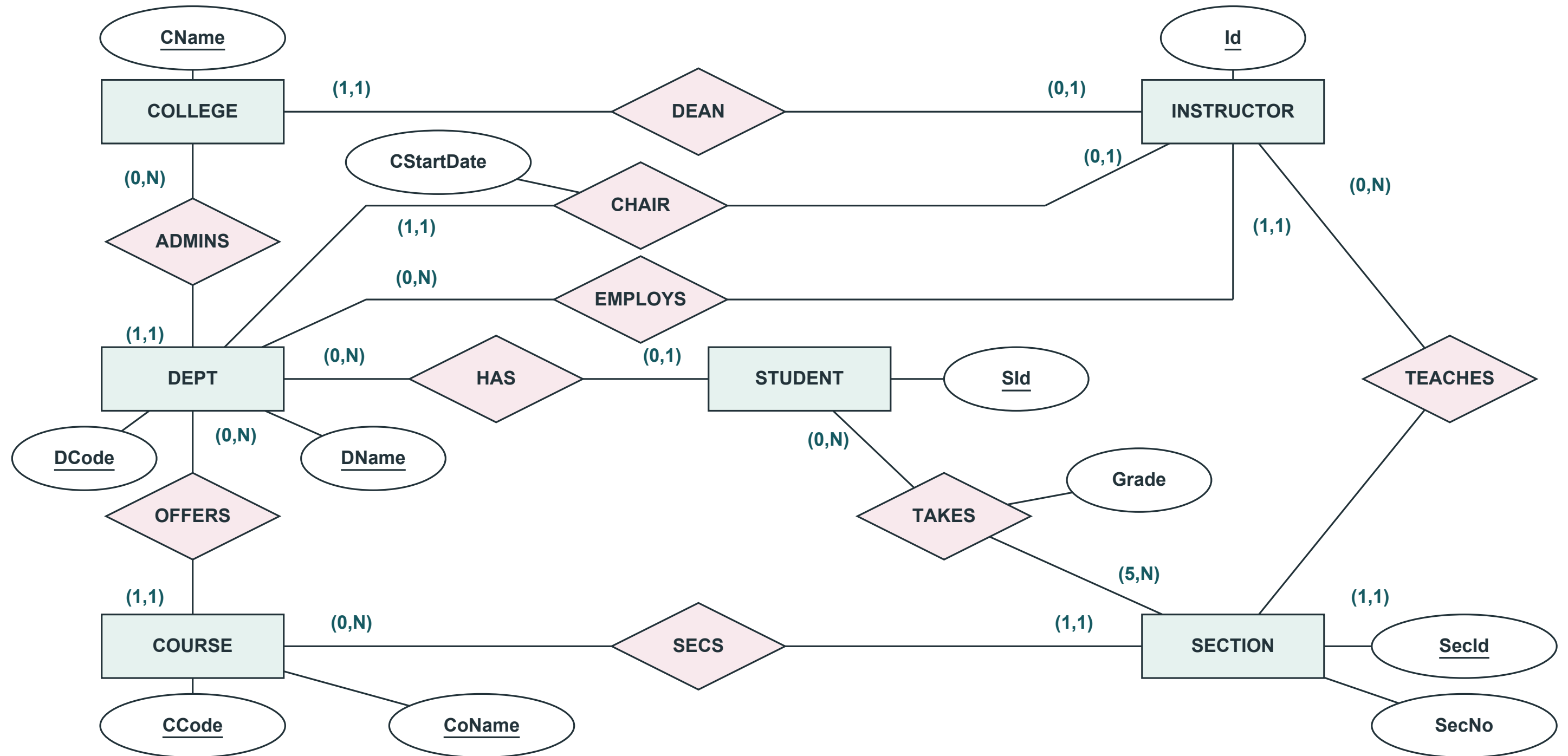


CORRECT: DB101 has Q101 in Fall 2026 and Q102 in Spring 2027. Both records remain.

Remember: Change COURSE's maximum from 1 to N, as in Figure 3.20. Each SECTION still belongs to exactly one course.

UNIVERSITY: The Corrected ERD

Reference answer: the relationships, keys and constraints in textbook Figure 3.20, redrawn after the six repairs.



All nine relationships, entity keys, CStartDate and Grade are shown. Other entity attributes are omitted for focus. HAS follows Figure 3.20, not the conflicting prose.

Remember: Explain the six changes using R1-R9 and the same sample records. The target is Figure 3.20, not a new university policy.

ERD Skills: Draw, Correct and Explain

Use the UNIVERSITY rules and records you just read. The same facts support all three kinds of ERD question.

Task	Concrete example
Draw from a description	R8: each section has one instructor; an instructor may teach none or many.
Correct an incorrect ERD	I1 teaches Q101 and Q102. Repair a diagram that limits I1 to one section.
Explain an ERD	INSTRUCTOR (0,N): zero or more sections. SECTION (1,1): exactly one instructor.

- 1. Draw:** connect INSTRUCTOR and SECTION through TEACHES.
- 2. Correct:** put (0,N) beside INSTRUCTOR and (1,1) beside SECTION.
- 3. Explain:** I1's two assignments are allowed; a section without a teacher is not.

Remember: State the rule, point to the records, then draw, repair or explain the matching symbols.

Chapter 3: Key Ideas

Use these questions to explain a diagram without reading long paragraphs.

Ask	Point to or explain
What object and what type?	One instance versus the type rectangle
What identifies it?	Complete key, or owner plus partial key
What is connected?	Relationship and named roles
How many may or must connect?	Maximum and minimum, separately
Whose property is this?	Entity attribute or relationship attribute
Which rules changed?	Small campus versus UNIVERSITY

1. Use the same small data when comparing two interpretations.
2. For APPROVES, name all three participants. Read a ternary 1 by fixing the other two.
3. SQL and relational mapping come later; neither is needed to explain these ER drawings.

Remember: Ch4 extends ER with shared properties and more specialized entity types.

Chapter 3: Textbook References

Elmasri and Navathe, Fundamentals of Database Systems, 7th Edition, Pearson.

Topic	Sections	Photographed pages
Requirements and conceptual design	3.1-3.2	90-93
Entities, attributes, keys, domains	3.3	93-102
Relationships, roles, constraints	3.4	102-108
Weak entities; min-max notation	3.5; 3.7.4	109-115 (selected)
Ternary facts and the meaning of 1	3.9.1-3.9.2 (selected)	118-122
UNIVERSITY example; Figure 3.20	3.10	122-124

1. Original English explanations, diagrams and illustrative data adapt the existing course examples.
2. Section 3.8 UML is not assigned in this chapter.
3. The UNIVERSITY prose/figure difference is retained explicitly. This teaching PDF is not a publisher's slide deck.

Remember: Use section numbers when another printing has different page numbers.